

Sudhanshu Srivastava

Louisville, KY • sudhanshu.iit.bit@gmail.com • +1 (502) 608-9718 • [LinkedIn](#) • [GitHub](#) • [Portfolio](#)

EXPERIENCE

Research Scientist

Aug 2024 – Present

IBM - UofL | Louisville, KY

- Led development of **CF-MARL**, a multi-agent causal **RL** framework for semiconductor defect root-cause analysis on 5,173 real-world records; reduced structural Hamming distance by 62.5% over 8 SOTA baselines, achieving 96.1% downstream prediction accuracy
- Designed **LHAC**, a lexicographic dual-agent deep RL production scheduler; achieved 98.5% completion at **8,340x faster inference** than ILP, outperforming GA+CPLEX by up to 53 pp under stochastic disruptions with real-time monitoring
- Developed **StressNav**, an RL-based navigation system in collaboration with **Gresham Smith** fusing 16 wearable **biomarker signals** and environmental features via **ETL pipelines**; reduced harmful stress by 4.6% for 94% of 68 participants (3-Minute Thesis Runner-up)

AI Consultant

Feb 2024 – Jun 2024

Hope and Dreams Trust | India

- Built **NLP-driven predictive analytics** pipelines for scholarship targeting across 1,000+ beneficiaries; developed automated reporting tools reducing manual data processing time by 40%, delivering **executive reports** and projections

Senior AI Research Associate

Aug 2023 – Dec 2023

IIM Ahmedabad | India

- Built end-to-end **ML/DL** pipelines for pricing optimization and demand forecasting on live transactional data for Nykaa (\$7B+ market cap); engineered customer segmentation on 500K+ records using **clustering and classification**, improving campaign targeting

Assistant Professor, Computer Science

Aug 2019 – July 2023

UPES Dehradun | India (on Sabbatical leave)

- Designed and delivered **ML, Deep Learning, and NLP** curricula to 400+ students across 8 semesters; supervised 8 capstone projects (2 peer-reviewed publications); filed 3 Government of India software copyrights for original ML tools

System Analyst

Jun 2016 – Jul 2017

Maven General Contracting L.L.C. | Abu Dhabi, UAE

- Built **data-driven forecasting models** for procurement and cost estimation across 10+ active sites; reduced cost variance by 18% and developed automated **analytics dashboards** improving project tracking efficiency

TECHNICAL SKILLS

- AI/ML**: LLMs, Generative AI, Agentic AI (LangChain, LangGraph), RAG, Reinforcement Learning (Multi-Agent, Hierarchical, PPO), Causal Discovery, GANs, Diffusion Models, NLP, Computer Vision, Transformers
- Languages & Tools**: Python, R, SQL, PyTorch, TensorFlow, scikit-learn, Stable-Baselines3, OpenAI Gym, pandas, NumPy, SciPy
- Infrastructure**: Docker, Git, Linux, FastAPI, MLflow, NVIDIA Triton, Kubernetes, AWS, Terraform, ETL Pipelines, Weights & Biases, CPLEX, Gurobi, Databricks

PROJECTS

End-to-End MLOps Platform | Feast, MLflow, Prefect, KServe, Evidently, FastAPI, Docker

[GitHub](#)

- Built a production **MLOps** platform with model registry, **drift detection**, automated retraining, **monitoring**, and **Kubernetes**-style serving pipelines

Low-Latency Triton Inference Stack | ONNX, TensorRT, NVIDIA Triton, Kubernetes, Prometheus

[GitHub](#)

- Developed a **high-performance model-serving** system reducing p99 latency from 41.2 ms to 6.8 ms using **Triton + TensorRT FP16** with HPA autoscaling and **Grafana monitoring**

Real-Time Voice RAG Agent | FastAPI, WebSockets, ChromaDB, OpenAI, AWS ECS, Terraform

[GitHub](#)

- Built an **agentic voice assistant** with faster-whisper, pyannote VAD, **ChromaDB RAG** retrieval, **LLM** support, and streaming TTS; deployed on **AWS ECS/Fargate** via Terraform

LHAC-DRL Manufacturing Scheduler | PPO, Stable-Baselines3, Streamlit

[GitHub](#)

- Published reference implementation of Lexicographic Hierarchical Actor-Critic for production scheduling with **PPO** training, baselines, evaluation scripts, and interactive **Streamlit** demo

Publications ([Click Here](#))

EDUCATION

University of Louisville - Louisville, KY

Aug 2024 – Dec 2026

PhD, Industrial & Systems Engineering | NSF-IBM Funded | **Thesis**: Reinforcement Learning for Distributed Manufacturing

Relevant Coursework – Generative AI, Predictive Analytics I, II, Operations Research

University of Texas at Austin - USA

Oct 2019 – Nov 2020

PGP, Artificial Intelligence & Machine Learning

Relevant Coursework – Machine Learning, Deep Learning, LLMs

UPES Dehradun - India

Jan 2023 – Jan 2025

MBA, Operations Management

Relevant Coursework – Operations Analytics, Service Operations, Production Planning & Inventory Control, Business Policy & Strategy

Indian Institute of Technology (IIT) Dhanbad - India

Jul 2017 – May 2019

MTech, Computer Science & Engineering

Relevant Coursework – Data Mining, Advanced Algorithms, High Performance Computing, Soft Computing, Cryptography

AWARDS & LEADERSHIP

TBE Fellow, Purdue (2026) | **Outstanding Service Award**, ISE Dept., University of Louisville — top 2 of all graduate students (2026) | **President**, IGSA, UofL (2025 - current) — led 9-member board — won **Outstanding New Org.** (2026) | **ASQ Scholarship** (2025) | **NSF Travel Award** (2025-26) | **Runner-up**, 3-Minute Thesis (2025)